

Building Damage Classification & Streamflow Bias Correction

CS 4440-101 · Artificial Intelligence · Final Project

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Task 1: Damage Classification

Task 2: Streamflow Bias Correction

Project Overview

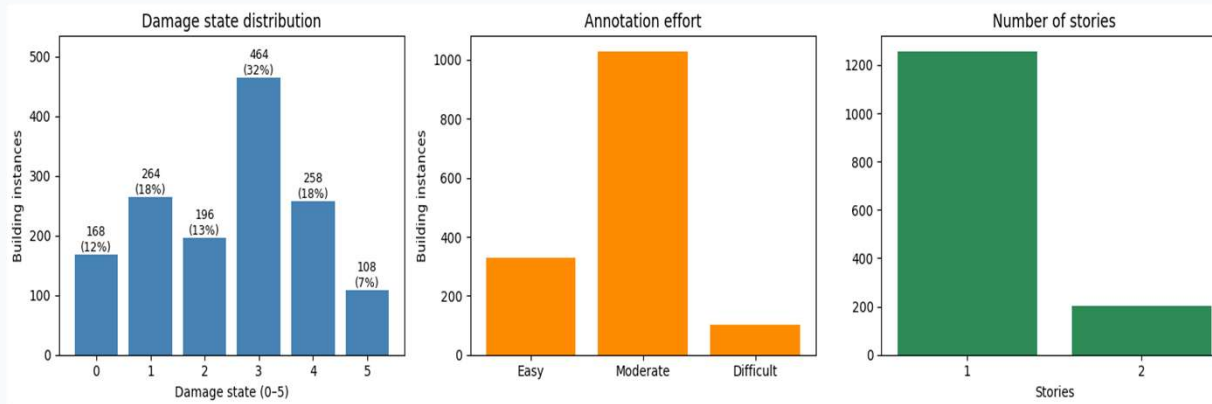
Task 1 · DoriaNET

- Post-hurricane building damage from UAV imagery
- 6-class ordinal classification (FEMA HAZUS 0–5)
- 1,458 instances · 147 physical buildings
- Building-stratified split (no leakage)
- Models: MLP · CNN · MobileNetV2

Task 2 · Runoff Forecasting

- Correct NOAA NWM streamflow forecasts at 2 stations
- Predict residual error; add back to NWM
- Lead times 1–18 hours · Apr 2021 – Apr 2023
- Chronological train/val/test split
- Models: GRU · LSTM simple · LSTM stacked

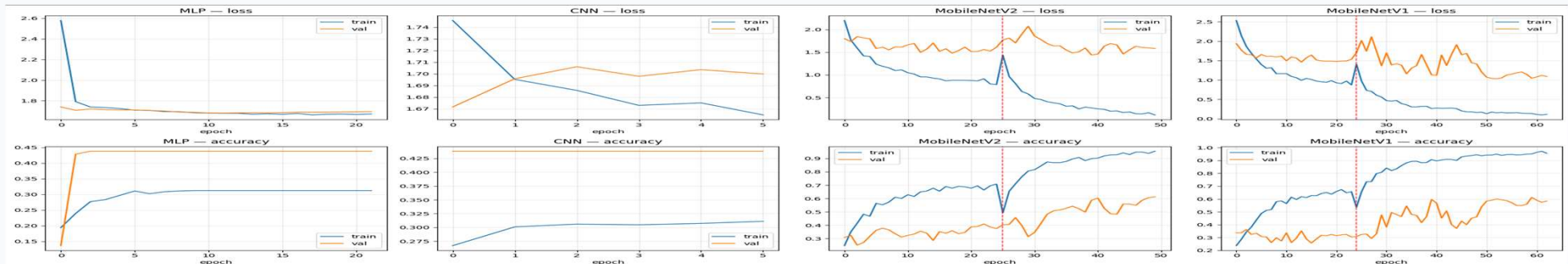
DoriaNET · Data & Pre-processing



1,458 instances 271 UAV frames, 147 physical buildings
Building-stratified split No building appears in 2 splits — prevents leakage
Augmentation Flip · $\pm 15^\circ$ rotation · $\pm 20\%$ zoom (train only)
Class imbalance Class 3 = 32%, class 5 = 7%; cross-entropy handles it
Split: 794 / 315 / 349 Train / Val / Test (seed=8 for balanced classes)

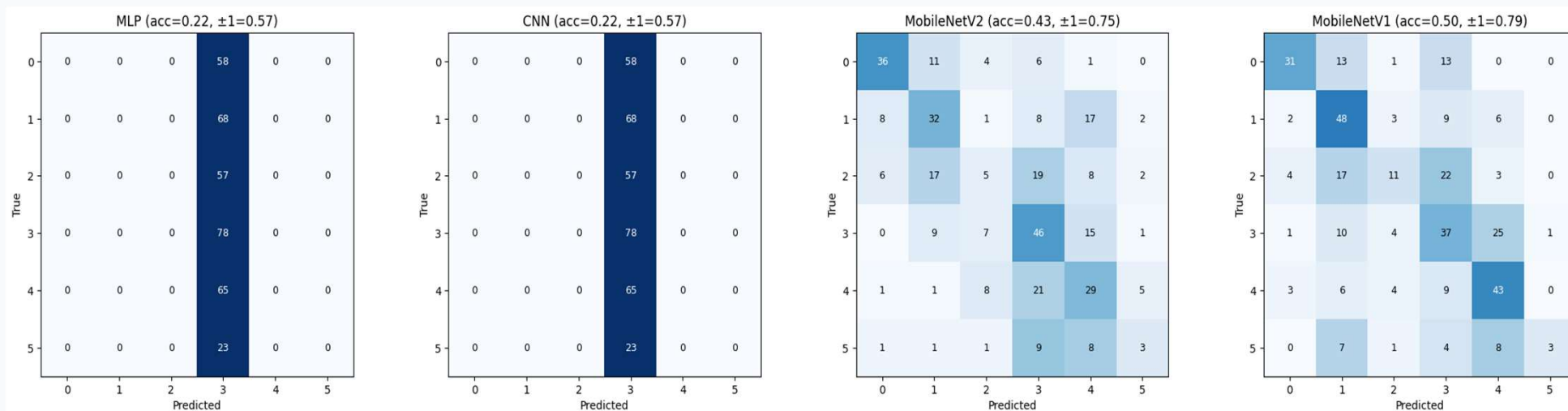
Key methodological choice: building-stratified split eliminates the data leakage present in Cheng et al. (2021). Expected accuracy: 30–50% exact (vs. paper's 61% with leakage) — matching paper's own cross-dataset baseline.

Training curves (all 4 models)



DoriaNET · Results

Confusion Matrices (test set)

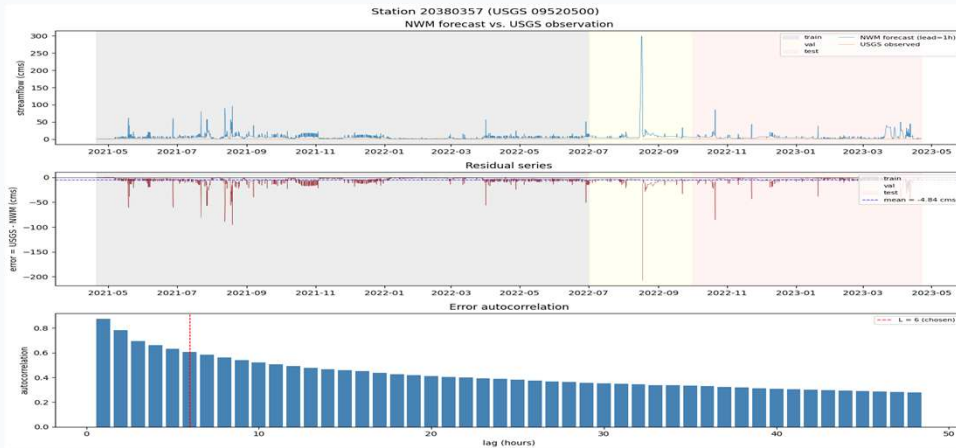


Model	Assigned To	Exact Acc.	± 1 Acc.	Val Acc.
MLP	Zach Shotwell	0.22	0.57	0.444
CNN	Miguel Moreno Coin	0.22	0.57	0.438
MobileNetV2	Alan Ray	0.43	0.75	0.613
MobileNetV1 ★	Alan Ray	0.50	0.79	0.613

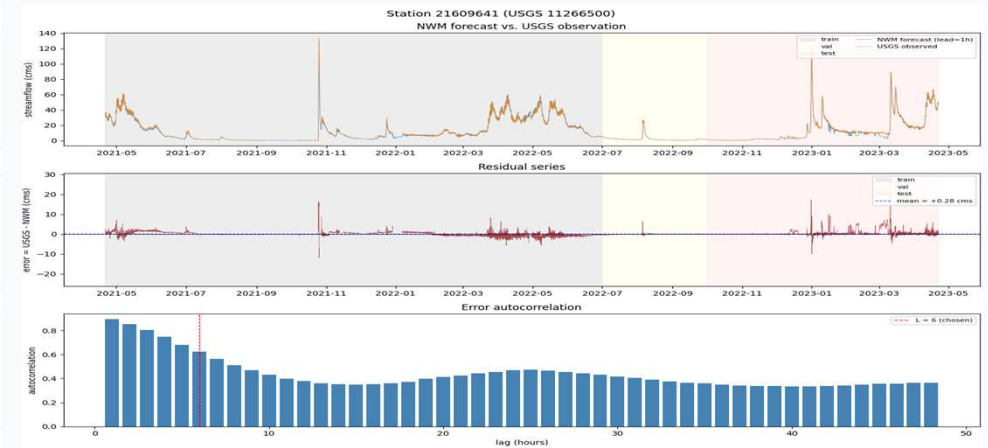
★ Best model · MobileNetV1 and V2 both benefit from ImageNet pretrained weights — critical on small datasets

Runoff Forecasting · Data & Pre-processing

Station 20380357 — Gila River, AZ (small, regulated)



Station 21609641 — San Joaquin River, CA (large, unregulated)



Pre-processing

- NWM format parsed with explicit strptime (~50× speedup)
- USGS resampled 15-min → hourly via asfreq
- Forward/back-fill missing values (0.4% / 5.8%)
- Chronological split: train Apr21–Jun22 · val Jul–Sep22 · test Oct22–Apr23

Feature Engineering

- Features: [error, NWM_q, USGS_q] at lead_h=1
- Window L=6h history → horizon H=18h
- Inputs z-scored on train only; targets raw cms
- L-sweep confirmed L=6 is optimal at station 1

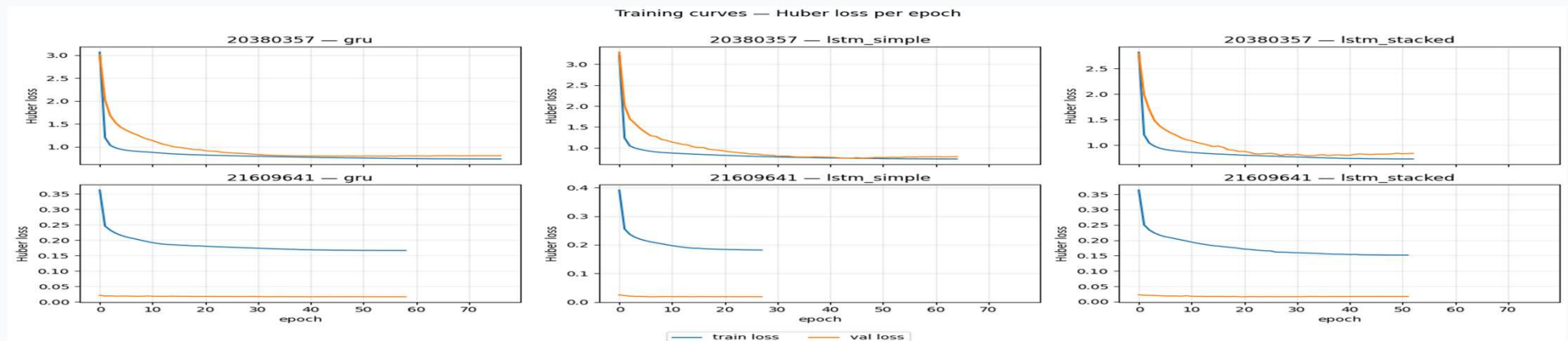
Station 1 (AZ): heavily regulated — Painted Rock Dam + irrigation diversions. NWM does not model dams → mechanically decoupled signals.

Runoff Forecasting · Models & Training

Model	Architecture	Params	Assigned To
GRU ★	GRU(32) → Dense(18)	4,146	Alan Ray
LSTM simple	LSTM(32) → Dense(18)	5,202	Zach Shotwell
LSTM stacked	LSTM(32, ret_seq) → LSTM(32) → Dense(18)	13,522	Miguel Moreno Coin

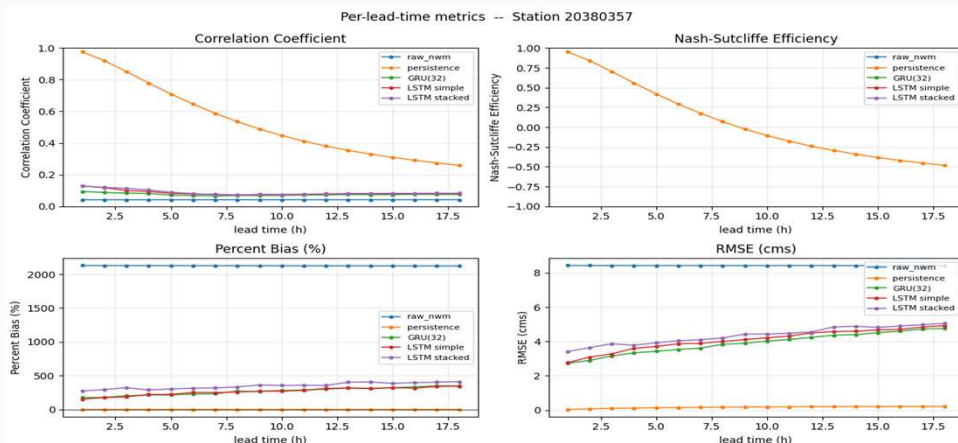
- Loss: Huber (robust to flood-peak outliers)
- Optimizer: Adam lr=1e-3, clipnorm=1.0 (BPTT exploding-gradient protection)
- Callbacks: EarlyStopping(patience=20) + ReduceLROnPlateau
- Hyperparameter search (GRU at sta. 21609641): units=64, lr=3e-3 → val MAE 0.0419 vs. 0.0447 for paper defaults

Training curves — Huber loss (both stations, all 3 models)

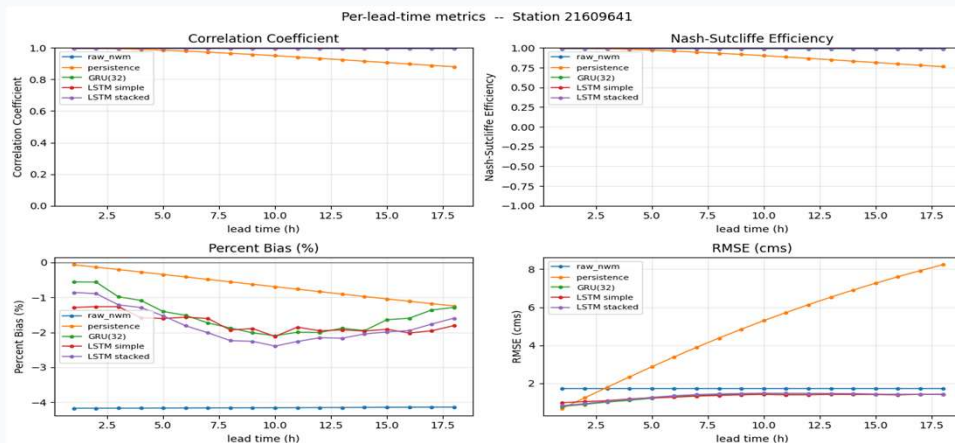


Runoff Forecasting · Results

Per-lead-time metrics · Station 20380357 (AZ)



Per-lead-time metrics · Station 21609641 (CA)

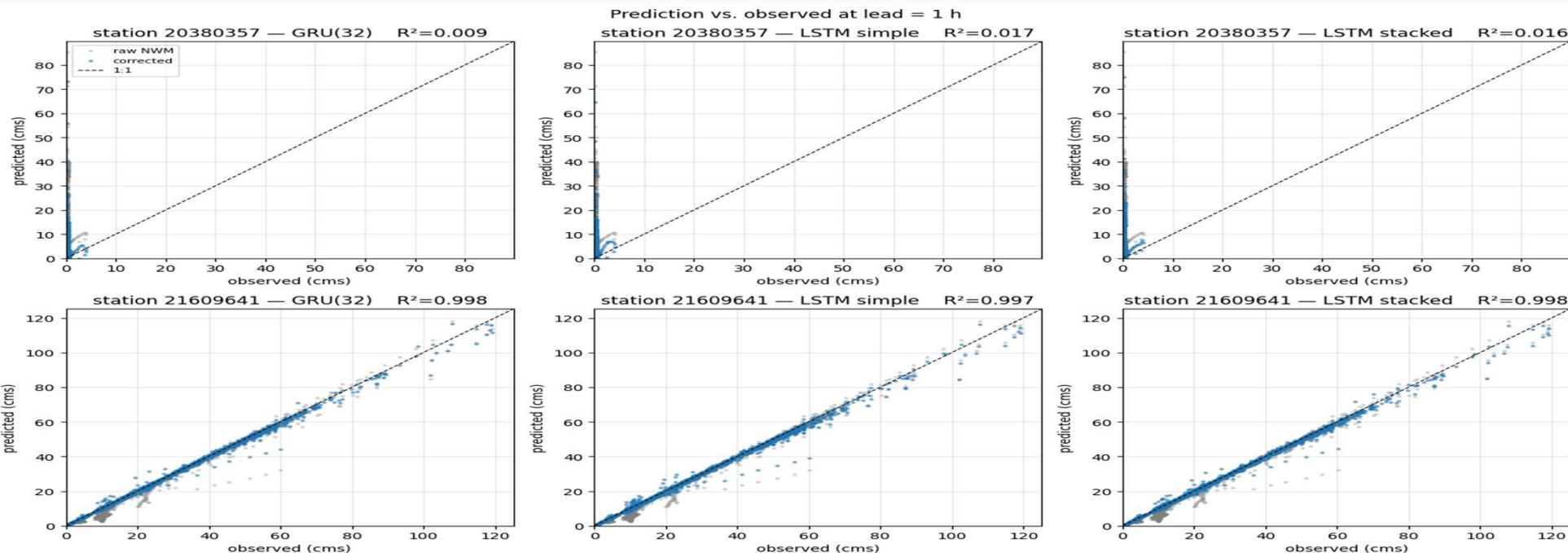


Station / Model	CC @ L1	NSE @ L1	PBIAS	RMSE L1 (cms)
20380357 Raw NWM	0.04	<<0	2130%	8.4
20380357 GRU ★	0.09	<0	165%	2.7
21609641 Raw NWM	0.995	0.989	-4.2%	1.73
21609641 GRU ★	0.999	0.997	-0.5%	0.86

Station 1 (AZ): post-processor cuts RMSE ×3 and PBIAS ×13 — but CC stays near 0 (regulation mismatch, not model failure) · Station 2 (CA): $R^2=0.998$ at all leads, extreme event generalization confirmed

Runoff Forecasting · Prediction vs. Observed

All models at lead = 1 h · GRU, LSTM simple, LSTM stacked (rows: station 20380357 top, 21609641 bottom)



Station 21609641 (CA) — Success

All three models achieve $R^2 \geq 0.997$ at lead=1h. Corrected forecast (blue) tightly clusters around the 1:1 line vs. scattered raw NWM (gray).

Station 20380357 (AZ) — Bias Correction Only

$R^2 \approx 0.01$ across all models. Corrected forecast moves mass from high NWM bias toward zero, but dynamic correlation is absent — expected given NWM–USGS decoupling.

Conclusions & Future Work

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- Transfer learning is essential — MLP & CNN collapse without pretrained features
- MobileNetV1 best: 50% exact / 79% ± 1 accuracy under leakage-free split
- Future: class-balanced sampling, ordinal loss (EMD^2), multi-location generalization

Runoff Forecasting

- Station 2 (CA): GRU halves NWM RMSE, $R^2=0.998$ through 18h including extreme events
- Station 1 (AZ): bias correction succeeds (RMSE $\times 3$, PBIAS $\times 13$) but dynamic skill absent — regulation mismatch
- Future: precipitation forcings, larger L to capture daily seasonality, Transformer encoder

Cross-Cutting

- Splitting choices matter: building-stratified & chronological splits give honest numbers
- Shallow models (GRU 4k params) outperform deeper ones under small-data regimes